

文献阅读笔记

面向自主制造的视觉—激光雷达移动机器人对接充电

文献信息 Feiyu Jia, Misha Afaq, Ben Ripka, Quamrul Huda, Rafiq Ahmad. "Vision- and Lidar-Based Autonomous Docking and Recharging of a Mobile Robot for Machine Tending in Autonomous Manufacturing Environments." *Applied Sciences*, 13(19), 10675, 2023. DOI: 10.3390/app131910675。

一句话概括 系统让 YOLOv7 从相机图像中回答“正确充电站在哪里”，再让 3D 激光雷达和七状态几何控制回答“怎样精确靠上去”；充电站检测在验证集达到 99.4% mAP@0.5，真实移动场景平均识别准确率为 95%。

研究动机

制造环境中的移动机器人要连续执行机床上下料等任务，必须自主寻找充电站并精确对接。单一传感器各有短板：相机信息丰富且便宜，但怕低照度、运动模糊和视场盲区；激光雷达测距精确、不依赖照明，但点云稀疏、成本和计算负担较高。作者因此用视觉做语义识别，用激光雷达做末端几何定位。

实验平台是 Clearpath Husky UGV，搭载 Hikvision 相机、Ouster 3D 激光雷达并运行 ROS Melodic；无线充电器安装在带侧壁的定制舱体内。

方法架构

流程分三部分：相机/雷达采集环境数据；YOLOv7 识别无线充电站；激光雷达根据墙面和充电器距离调整机器人姿态并完成靠站。相机内参和雷达坐标通过变换矩阵关联，可由检测框映射目标的 3D 坐标。

then dock it to the detected wireless charger. A flowchart of the proposed method is presented in Figure 3.

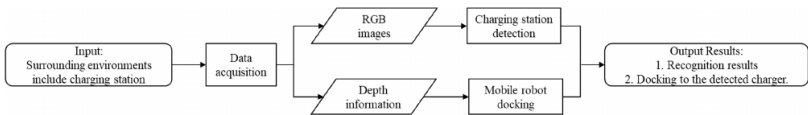


Figure 3. A block diagram of the proposed docking and recharging method.

4.1. YOLOv7 Architecture

YOLOv7 is a one-stage model and the latest algorithm for real-time object detection, and it performs well in terms of both speed and accuracy [26]. The architecture of the proposed charging station detection method based on YOLOv7 is presented in Figure 4, and it is composed of three main components: a backbone, neck, and head. The convolutional backbone module adopts Darknet-53 [27] to extract image feature maps from the input image and transfer them to the neck layers. In the neck module, the Feature Pyramid Network (FPN) [28] is used to enhance the feature maps. These maps are then combined, fused, and passed to the subsequent layers. Finally, the head network predicts the bounding boxes and classes of the objects.

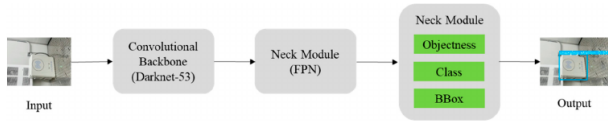


Figure 4. A flowchart of the charger detection method.

YOLOv7 adopts a developed extended efficient layer aggregation network to improve inference efficiency. This network can quicken the model’s learning ability without disturbing or changing the original gradient propagation path. In addition, a novel scaling method, referred to as corresponding compound model scaling, is proposed to address the issue of a larger width output of the computational block by directly scaling the depth of the concatenation-based model. Moreover, several techniques are used to improve inference accuracy while keeping training costs low. These techniques, called Bags of Freebies (BoF), include planned re-parameterization, dynamic label assignment, and batch normalization. After thoroughly investigating the re-parametrized convolution, the authors demonstrate increased model accuracy when using RepConv without an identity connection. Furthermore, batch normalization integrates the mean and variance in the data to adjust the bias and weight of the convolutional layer, which can immediately impact the training process by utilizing a higher training rate and faster convergence.

According to [26], YOLOv7 optimizes the inference process and improves detection accuracy and speed compared with other existing real-time object detection methods because of its more advanced network structure and training strategies. However, it has not yet been used in the domain of autonomous docking and recharging. In this article, YOLOv7 is adopted as the backbone architecture to detect and recognize the charging station.

图：原文第6页，给出整体框图与YOLOv7 充电站检测流程。

七状态激光雷达对接

点云被滤出四个距离量：前向斜线 `Front_laser`、后向斜线 `Back_laser`、侧墙 `Wall_laser` 和充电器 `Charger_laser`。控制状态机如下：

1. 比较前/后斜线距离，先把机器人调正。
2. 根据 `Wall_laser` 与已知目标距离判断机器人在充电器左侧还是右侧。
3. 向相应方向转 90° 附近，使一条斜线与墙距相等。
4. 直线横移到目标侧向位置。
5. 再次调正，使机器人与充电器正对。
6. 直行至距离充电器 2–3 cm。
7. 停车并把状态置为 `docked`。

```
Algorithm 1. Lidar-based docking.
State 1: Robot straightening
Initialize Front_laser, Back_laser, Charger_laser, and Wall_laser
If (Front_laser - Back_laser) > 0 then rotate clockwise until Front_laser = Back_laser
elseif (Front_laser - Back_laser) < 0 then rotate anti-clockwise until Front_laser = Back_laser
If Wall_laser > known_distance
Change state to 3
elseif Wall_laser < known_distance
Change state to 2
State 2: Robot turning left if to the right of the charger
Turn the robot anti-clockwise until Back_laser = Wall_laser
Then change state to 4
State 3: Robot turning right if to the left of the charger
Turn the robot clockwise until Front_laser = Wall_laser
Then change state to 4
State 4: Robot's linear motion
Move robot in a linear motion until Wall_laser = known_distance
Then change state to 5
State 5: Robot straightening second time
If (Front_laser - Back_laser) > 0 then rotate clockwise until Front_laser = Back_laser
elseif (Front_laser - Back_laser) < 0 then rotate anti-clockwise until Front_laser = Back_laser
Then change state to 6
State 6: Robot moving towards the charger
Move robot in a linear motion until charger_laser within 2 to 3 cm's away from the charger
Then change state to 7
State 7: Robot docking with the charger
Stop the robot's motion and change status to docked
```

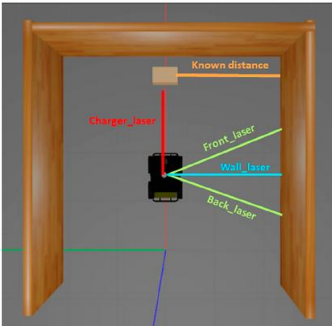


Figure 7. Lidar-based docking method visualization.

The above algorithm was tested on the Husky robot and gave a fairly accurate pose estimation and localization for docking, except for systematic errors based on the Lidar data readings. The algorithm was tested for various initial poses and locations from the charger; a few of the scenarios considered can be seen in Figure 8. The experiments conducted on the actual Husky robot are shown in Figure 9. The known distance of the charger from the side wall can be determined using the vision-based method and matched with the Wall_laser to fuse the Lidar-vision data. Moreover, once the robot is in the correct docking position for charging or close to the desired location, the Lidar point cloud data and the

图：原文第9页，状态机把复杂的二维位姿修正拆为可解释的旋转和平移。

数据集与模型训练

由于没有公开的同类充电站数据集，作者采集 240 张 1920×1018 图像：160 张训练、40 张验证、40 张测试，使用 Labellmg 标注。数据增强包括旋转、平移、缩放、垂直翻转、Gaussian 模糊和噪声。模型基于 COCO 预训练权重做迁移学习。

训练硬件为 RTX 3090、Threadripper 3970X 和 128 GB RAM；学习率 0.001、动量 0.9、batch size 16，并比较 100–300 以上 epoch。约 300 epoch 时训练/验证损失稳定；再继续训练，验证损失回升，出现过拟合。

结果

指标	结果
验证集 mAP@0.5	99.4%
验证集 mAP@0.5:0.95	86.5%
真实移动视频平均识别准确率	95%
末端停止距离	充电器前 2–3 cm（算法设定）

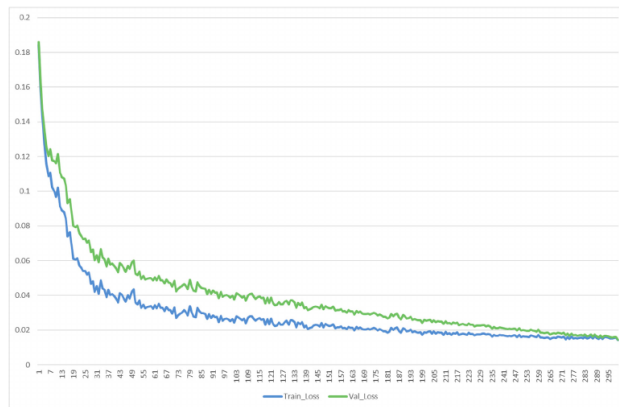


Figure 11. Training loss and validation loss of the charger detection model.

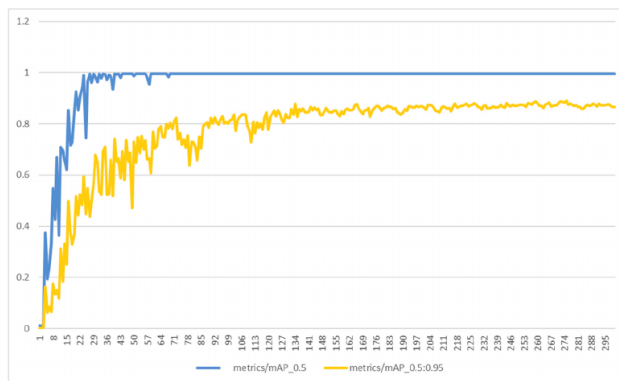


Figure 12. The results of both performance metrics.

图：原文第 13 页，训练损失在约 300 epoch 收敛，并给出检测性能。

贡献如何评价

论文的合理分工是最大亮点：视觉负责在多个外观相近目标中选择正确充电站，点云负责精确距离与姿态控制。七状态控制器规则透明，易于在 ROS 中调试，比端到端策略更容易定位失败原因。

与 2005 年的视觉地标对接论文相比，本研究不要求固定色彩图案，目标识别更灵活；但末端几何控制仍依赖人工设计的充电舱侧壁，说明“深度学习识别”没有消除环境工程化要求。

局限与批判性阅读

- 数据集仅 240 张、单类目标且来自特定场景，99.4% mAP 不代表跨工厂、跨相机或跨充电器泛化。
- 文中报告了识别准确率，却没有系统给出端到端对接成功率、对接耗时、初始位姿工作域和重复试验统计。
- 方法假设充电站位于封闭空间并有可测侧墙；开放式站点或动态遮挡会破坏 `Wall_laser` 逻辑。
- 论文称融合可降低计算成本，但缺少与纯点云、纯视觉方案的延迟、算力和能耗对照。
- 低照度和运动模糊仍会影响检测；相机—雷达标定在论文中更像方法框架，作者也把更完整标定列为后续工作。

可复现与改进建议

复现时应先做三组基线：仅视觉、仅雷达、视觉+雷达；每组在统一的初始距离/角度网格上至少重复若干次，报告成功率、耗时、最终横向/角度误差和计算延迟。工程升级可加入 AprilTag 或主动红外作为低照度备份，采用自动外参标定、鲁棒平面拟合和接触/充电电流确认，最终把“检测成功”提升为“真正建立稳定充电连接”。

来源与使用说明

本笔记依据本地保存的原论文整理；图片均为原文关键页面的本地渲染，仅用于团队内部学习、讨论和设计参考。笔记中的评价与启发属于阅读分析，不等同于作者原文结论。